INCH-POUND
MIL-PRF-3098/116E
11 July 2011
SUPERSEDING
MIL-PRF-3098/116D
27 July 2005

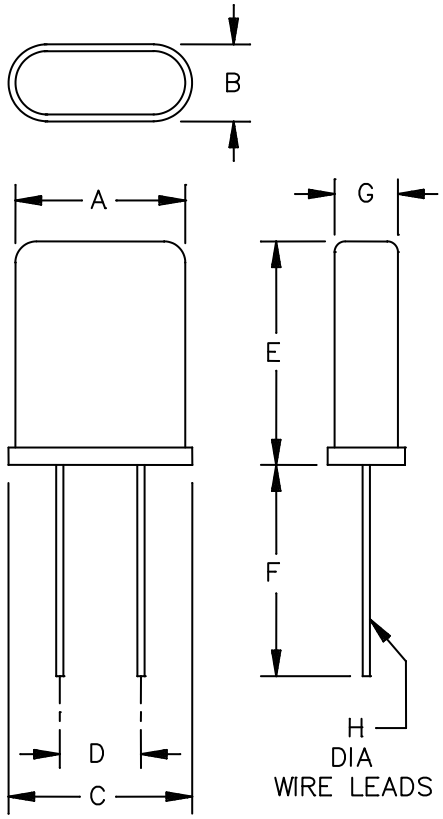
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR137/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 8.5 MHz to 19.0 MHz; fundamental; noncontrolled; antiresonance.



Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.402	---	10.21
B	---	.183	---	4.65
C	---	.435	---	11.05
D	.184	.200	4.67	5.08
E	---	.530	---	13.46
F	.50	---	12.7	---
G	---	.150	---	3.81
H	.015	.019	0.38	0.48

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR137/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 8.5 MHz to 19.0 MHz, inclusive.

Frequency tolerance:

Operating temperature range: Reference temperature frequency ±25 ppm.

Reference temperature: ±25 ppm.

Equivalent resistance:

Frequency range inclusive	Maximum resistance
8.5+ to 9.99999 MHz	35 ohms
10.0 to 19.0 MHz	25 ohms

Mode of oscillation: Fundamental.

Reference temperature: 40°C ±2°C.

Rated drive level: 1.0 mW, maximum.

Capacitance, shunt: 7.0 pF, maximum.

Antiresonance load capacitance: 32.0 pF ±0.5 pF.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Part or Identifying Number (PIN): For Product Level B, PIN shall be CR137/U or CR137B/U (followed by specified frequency). For Product Level S, PIN shall be CR137S/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-PRF-202

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review activities:
Air Force - 19, 84

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(Project 5955-2011-006)

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